

Finding Equilibrium in Multi-Agent Games with Payoff Uncertainty



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Two-player zero-sum game: Rock-Paper-Scissors

- Player 1 and Player 2
- Zero sum:

WIN: Player 1 gets 1 point, Player 2 loses 1 point

LOSE: Player 1 loses 1 point, Player 2 gets 1 point

- Pure strategy: e.g. always plays Rock
- Mixed strategy: $x = (x_1, x_2, x_3) \in \Delta$
 $y = (y_1, y_2, y_3) \in \Delta$

Nash Equilibrium: A pair of strategies (x^*, y^*) , s.t.

$$x^* \in \arg \max_{x \in \Delta} x^T A y^* \quad y^* \in \arg \max_{y \in \Delta} (x^*)^T (-A) y$$

	$\frac{1}{3}$ 	$\frac{1}{3}$ 	$\frac{1}{3}$ 
$\frac{1}{3}$ 	0	-1	1
$\frac{1}{3}$ 	1	0	-1
$\frac{1}{3}$ 	-1	1	0

Payoff matrix A for Player 1.

Games with incomplete information

- With **complete game information**:
 - Nash equilibria for **general-sum** finite games can be intractable to compute
 - Nash equilibrium of a two-player **zero-sum** game can be computed efficiently
- In practice, the parameters of a game are often only **partially known**
 - Companies get uncertain profit
 - Bidders have uncertain gain ...etc

How can we find **equilibria** in games in a way that is **robust to uncertainty** in the game parameters?

This work: payoff uncertainty

Two-player zero-sum game: Rock-Paper-Scissors

- Player 1 and Player 2
- Zero-sum
- Uncertain payoff matrix: $A \in \mathcal{U}$

If Player 1 wins: win $\in [0, 1] \sim$ uncertainty set

If Player 1 loses: loss $\in [-1, 0] \sim$ uncertainty set

			
	0	[-1,0]	[0,1]
	[0,1]	0	[-1,0]
	[-1,0]	[0,1]	0

Uncertain payoffs for Player 1.

Goal: Nash Equilibria for both players remains to be a Nash for any feasible payoff.

If such strategies exist, they are known as ***ex-post equilibrium***

Ex-post equilibrium

Suppose $A \in \mathcal{U}$ where $\mathcal{U}$ is some uncertainty set.

A pair of strategies x^*, y^* is at an **ex-post equilibrium** if:

$$\begin{aligned} x^* &\in \arg \max_{x \in \Delta_n} x^T A y^* \\ y^* &\in \arg \max_{y \in \Delta_m} x^{*T} (-A) y \end{aligned} \quad \forall A \in \mathcal{U}$$

In this work: (“distribution-free” uncertainty) We consider uncertainty sets that correspond to convex combinations of a set of known matrices:

$$A \in \mathcal{U} = \mathbf{conv}(\{A_1, \dots, A_k\})$$

Contribution of this work

- **In zero-sum one-shot games:**

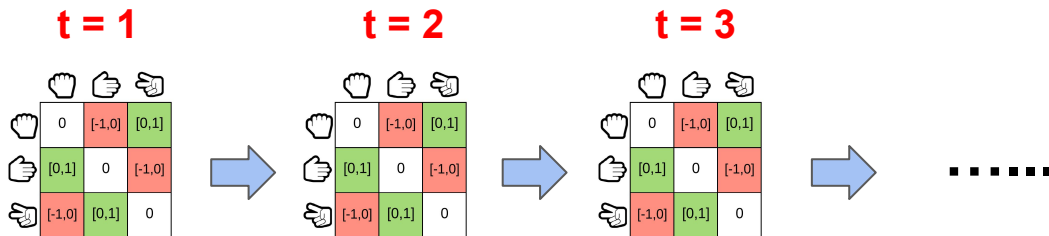
- Existing ex-post equilibrium of zero-sum polymatrix games can be found efficiently using linear programming

A 3x3 payoff matrix for a zero-sum Rock-Paper-Scissors game. The columns represent Player 1's strategy (Rock, Paper, Scissors) and the rows represent Player 2's strategy (Rock, Paper, Scissors). The payoffs are: (Rock, Rock) = 0, (Rock, Paper) = -1, (Rock, Scissors) = 1, (Paper, Rock) = 1, (Paper, Paper) = 0, (Paper, Scissors) = -1, (Scissors, Rock) = -1, (Scissors, Paper) = 1, (Scissors, Scissors) = 0. The matrix is color-coded: green for positive payoffs, red for negative payoffs, and white for zero payoffs.

	Rock	Paper	Scissors
Rock	0	-1	1
Paper	1	0	-1
Scissors	-1	1	0

- **In repeated stochastic games:**

- Ex-post equilibrium $\rightarrow$ Ex-post Markov perfect equilibrium (MPE)
- Sufficient conditions for the existence of such an equilibrium
- Any feasible values of a two-player zero-sum stochastic game can be bounded in a tight interval using dynamic programming



One-shot game: Existence of Ex-post equilibrium

- **Ex-post equilibrium doesn't always exist**
 - It exists only if the set of Nash Equilibria for all possible payoff matrices contains a common strategy.
- **Example:**
 - Having uncertainty in the upper right corner would mean that Player 1 would have a bias against playing Rock, thus the equiprobable strategy is no longer at Equilibrium.

			
	0	-1	[0, 1]
	1	0	-1
	-1	1	0

Payoffs for Player 1.

Result: Any existing ex-post equilibrium can be found via Linear Programming for certain zero-sum games with bounded polyhedral uncertainty sets.



Example: ex-post equilibrium exists

As a concrete example, if for all payoff matrices in the uncertainty set, the payoffs for winning and losing in are all equal in magnitude, then $x^* = y^* = (1/3, 1/3, 1/3)$ is an ex-post equilibrium.

$$\mathcal{U} = \text{conv}(\{$$

			
	0	-1	1
	1	0	-1
	-1	1	0

$$,$$

			
	0	0	1
	1	0	0
	0	1	0

$$\})$$

Example: ex-post equilibrium does not exist

If we are only uncertain about how much Player 1 would win in a Rock-Scissor setting, having $W < 1$ would mean that Player 2 would have a bias for playing Rock, thus the equiprobable strategy is no longer at Equilibrium.

$$\mathcal{U} = \text{conv}(\left\{ \begin{array}{c|cc} & \text{Rock} & \text{Paper} & \text{Scissors} \\ \hline \text{Rock} & 0 & -1 & 1 \\ \text{Paper} & 1 & 0 & -1 \\ \text{Scissors} & -1 & 1 & 0 \end{array} \right\}, \left\{ \begin{array}{c|cc} & \text{Rock} & \text{Paper} & \text{Scissors} \\ \hline \text{Rock} & 0 & -1 & 0 \\ \text{Paper} & 1 & 0 & -1 \\ \text{Scissors} & -1 & 1 & 0 \end{array} \right\} \right)$$

Further: We provide characterizations of the **maximal uncertainty sets** for an ex-post equilibrium to exist. (*details in the paper*)

Solving for existing ex-post equilibrium

LP(1)

$$\begin{aligned} \min_{\substack{x,y \\ w_{\{x,1\}}, \dots, w_{\{x,k\}} \\ w_{\{y,1\}}, \dots, w_{\{y,k\}}}} \quad & \sum_{i=1}^k w_{\{x,i\}} + w_{\{y,i\}} \\ \text{s.t.} \quad & w_{\{x,i\}} \geq -e_j^T A_i y \quad \forall j = 1, \dots, n; i = 1, \dots, k \\ & w_{\{y,i\}} \geq x^T A_i e_l \quad \forall l = 1, \dots, m; i = 1, \dots, k \\ & x^T \mathbf{1} = 1, x \geq 0 \\ & y^T \mathbf{1} = 1, y \geq 0 \end{aligned}$$

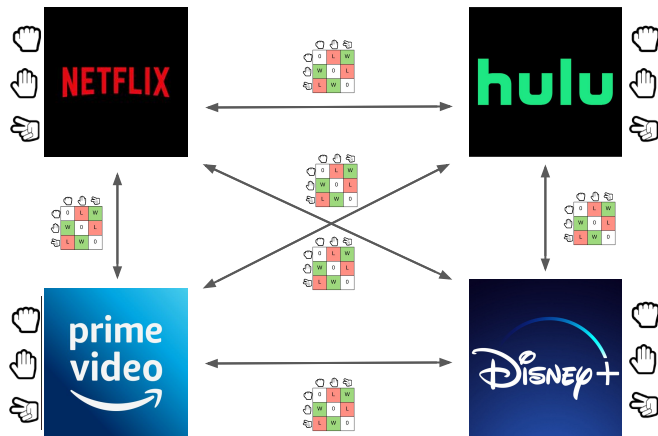
Theorem 1 (simplified two-player case)

Any ex-post equilibrium (x^*, y^*) for a two-player zero sum with uncertain payoff matrix $A \in \mathcal{U} = \text{conv}(\{A_1, \dots, A_k\})$ gives an optimal solution to LP(1).

Conversely, any optimal solution to LP(1) is an ex-post equilibrium of the game.

Extending to multi-players

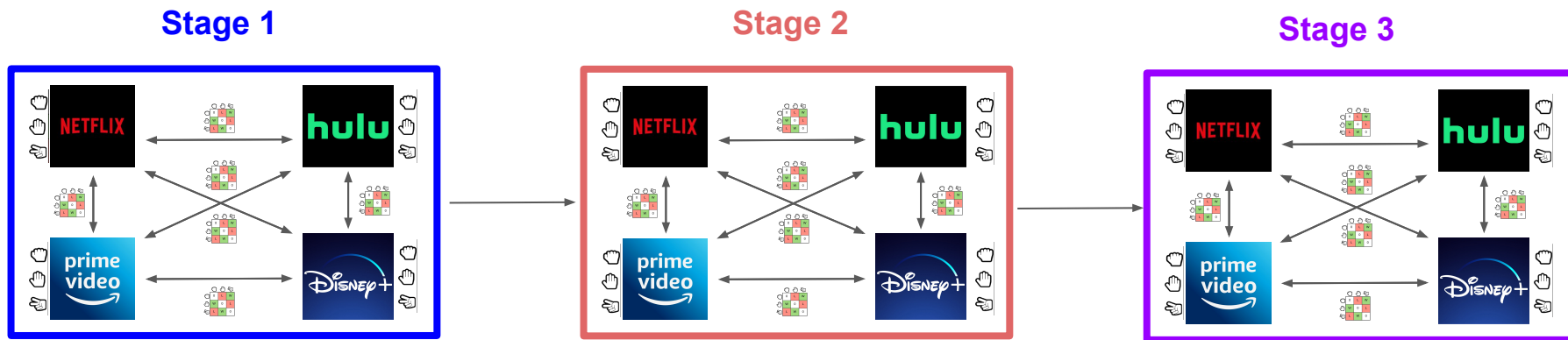
Theorem 1 holds for more general N-player **polymatrix games**:



- **Each pair of players** play a two-player matrix game
- Each player chooses **a single strategy**, which is applied to all his pairwise games
- **Zero-sum**: The sum of the payoffs for all players = 0

Extending to stochastic games with infinite time horizon

Stochastic game: Game is played over and over. Each stage has a **stage payoff matrices**, with transition probability between stages given by $P(\text{new stage} \mid \text{current stage, players' actions at current stage})$.



Example: the streaming services' actions in **Stage 1** change user behavior in **Stage 2**.

Ex-post Markov perfect equilibrium (simplified two-player case)

Player 1's expected discounted payoff: (2 players with strategies x, y)

$$\Pi_1(x, y; s^0) = \mathbb{E} \left[\underbrace{\sum_{t=0}^{\infty}}_{\text{discount factor}} \underbrace{\gamma^t}_{\text{discount factor}} \underbrace{M_1(x(s^t, \mathcal{H}^t), y(s^t, \mathcal{H}^t); s^t)}_{\text{payoff at stage } s^t} \right]$$

initial stage *discount factor* *payoff at stage s^t*

Ex-post MPE: A set of strategies $(x^*(s), y^*(s))$ is an ex-post MPE of the stochastic game iff it maximizes the expected discounted payoff for all the players under all feasible stage payoff matrices $A(s) \in \mathcal{U}(s), \forall s \in \mathcal{S}$.

Results: (applies to N-player)

- **Sufficient conditions** for the existence of such an ex-post MPE;
- **Value of the game** is achieved at MPE. It can be **bounded up to a tight interval** using dynamic programming

Future directions

- **Necessary conditions** for the existence of ex-post equilibria, and ex-post MPE
- Games with **more general payoff structures** (beyond polymatrix) and **uncertainty sets** (beyond polyhedral)
- Uncertain payoffs: **potential relations to practical reinforcement learning** problems with proxy rewards
- **Other sources of uncertainties** beyond payoff: transition probabilities

For more details, please see the paper at:

<https://arxiv.org/pdf/2007.05647.pdf>

Thank you for your time!

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